Wet Dream

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When first approached to write an article for *BC Caver magazine*, my first thought was "I'm not really an explorer, what would I have to share?" While I have participated in exploration, it is not my primary driver. I'm also not a real caver. To be honest, dry caving in small spots scares me. I much prefer flooded caves.

I've slowly realized over the years that I primarily dive to see new and interesting things. And in order to share them, photography has become my tool of choice. I started taking images as a child with old film cameras from the 50s. You would look in the top, take an exposure, and wind the film. Eight shots were all you got. My cave training took place down in the warm caves of Quintana Roo, Mexico. I was hooked but knew nothing of the local scene.

I first met Peter Norris at a shipwrecks convention which took place at Simon Fraser University. Marc-Andre Bernier, Chief of Parks Canada Underwater Archaeology Service, had just returned from the Arctic after discovering the wreck of the HMS Erebus. The Erebus was one of two lost ships from Sir John Franklin's 1845 expedition. I just happened to sit at the same table as him, and our topic of discussion soon turned to cave diving.

Peter told me about an exploration that he had just finished: Wet Dream Cave. The cave is located on Vancouver Island on the way to Zeballos. It's the second longest underwater cave in Canada and quite warm by local standards. The reason that it's warmer than others, is that the water flows in a shallow river bed prior to disappearing into a swallet. The resurgence is the only entry point. Any rain can quickly make the river water rise and make entry impossible. This means that it's not possible to dive the cave for most of the year, only in favorable dry weather conditions.



The thought of diving a local cave was intriguing and hung in my mind all winter. I had a rough map showing passage depths and directions on how to get there on the logging roads. Late summer was the best bet for clear skies. A good friend and fellow cave diver, Martin Pelletier, joined me for the trip. In late August of 2016, we loaded about 20 scuba tanks, dive, and camping gear into a large van, and headed to the Island from Vancouver. We arrived at where we thought the cave was in late afternoon, scrambled down to the dry riverbed and walked in what we hoped was the right direction. After a few hundred metres, we turned a corner and saw water flowing into the river from under a solid piece of limestone. We made it!

Over the next two days, we did three dives. The first dive was to have a look and get to know the cave. The first thousand feet is relatively shallow with large passage. Most of this section is tubular with walls worn away from the high flow and river rock. Some areas are quite marbled with lighter limestone and iron-rich veins. There are breakdowns with rectangular and triangular shards which have fallen from the ceiling. The bottom of the cave varies from patches of fine sand, smooth limestone, to hard packed gravel. Chunks of harder rock emerge from the floor and walls.

My camera at the time was a Canon G15, sealed in a Fantasea housing. The housing fixed down onto a tray, which is a metal bar with handles. Attached to the handles with ball joints are arms, which fix to two strobes. On the front of the housing was a wide-angle wet lens converter. The Canon G15 is a high end, fully manual, point and shoot camera. It does not handle low light particularly well. I quickly realized that for the size of passage and dark tone of rock, I needed a camera with much better sensor sensitivity and more light! I will discuss all aspects of setting up an underwater photography shoot in Wet Dream Cave: safety, logistics, lighting, and modelling.

Before taking a camera in any cave, it is best to gain experience diving in the cave. By doing dives with no camera, you can get to know the layout, lines, and scout shoot locations. Wet Dream Cave is shallow for the first three hundred metres and does not have many side passages, so is not a complex cave if you stick to the front. It is not particularly silty, so the danger of a silt-out is quite low. There can be current depending on rain events. After diving the cave in different conditions, you can get a sense of gas consumption and decompression obligations for specific locations. When diving, the rule of thirds (one third in, one third out, and one third contingency). I always want to know how much gas it will take for me to exit from the shoot location while conserving a third of my gas. I don't take pictures on the way in.

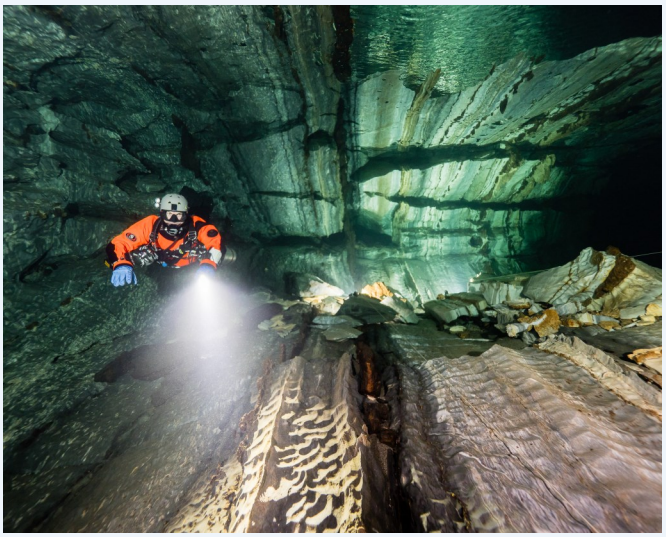
Breath taking view - Photo credit Eli Wolpin

Once the furthest shoot location is reached, I note the amount of gas used, and add that to the contingency to get the absolute turn pressure, though often turn before then to capture a few locations on the way to exit.

Bright lights and changing positions can be disorienting. You can also get drawn into the camera and lose focus of what is around you. It is often good practice to put a REM (referencing exit marker) or an arrow on the line prior to shooting so that in the event you become disoriented, you can be reminded of the exit direction.

The trip to Wet Dream Cave is a ferry ride and a few hours on the road from Vancouver to Vancouver Island. It is remote, so limited access to grocery stores or tank fills. You can solve this in a variety of ways on open circuit: a compressor, or lots of tanks. Bringing a lot of tanks is heavy, but you avoid having to run a compressor in the evenings. Camping gear and a generator to charge lights rounds out the general logistics.

I always bring spares for my camera gear. You never know if a fiber optic cable will snap leaving you one less strobe. Prior to the dive, I change all camera batteries, clean and lubricate all seals and assemble my camera inside the housing. My current camera is an Olympus OMD-EM1 mk2. This format is micro 4/3rds, which gets you a smaller camera but sacrifices on sensor size. Once my strobes and camera are all assembled, I test fire each strobe independently. There is nothing worse than being a half hour into a dive only to realize later you've put a battery in upside down.



No bubbles - Photo credit Eli Wolpin



Setting up the shot - Photo credit Eli Wolpin

Before the dive, all dive gear is ready and next to the entrance. Having assistance prior to the dive to hand you the camera and remote strobes, is always appreciated. I leave the neoprene glass dome protector on to ward off damage on the trip into the cave. I also clip together the camera arms so that they are tight and cannot move. Once I arrive at the photo shoot location, I remove the clip on the arms and the neoprene protector.

In caves, you make your own light. There are two types of lights which you can use underwater: strobes, and still (video) lights. Both sources have their benefits and drawbacks. Strobes produce a burst of light so are generally brighter and more adjustable. Video lights typically have only a few settings and typically produce lower quality light, though you can see the effect and hide them with ease.

When shooting a model in a cave, there are a few different lights to consider: onboard strobes, focus lights, remote lights, and the primary dive light of the model.

For onboard lighting, the focus lights are there just to allow the camera to focus on the subject. They are set low and I also use them to aid in communication. The onboard strobes can be tuned up or down depending on what needs to be lit. They will overpower the focus lights, so interference is generally not a concern. I position mine on long arms (metal extensions from the camera), generally behind the lens position.

The first few shots I take are to understand the way light works in the section of cave and to balance the primary light of the model. I always like to see the primary dive light of the model as a bright beam. It adds dynamic quality to the photographic composition. I tune my strobes up or down to check that I am not overpowering their light.

A brighter primary light for the model is always preferred, it means that I have more latitude with lighting so can use a higher f-stop, lower ISO, and faster shutter speed.

I try as much as possible to hide the light sources. This places emphasis on the formations being lit and not on the method. To do this, I tuck the strobe behind a rock and only leave the sensor exposed enough to trigger. Another method is to mount a remote strobe onto the model. The sensor can be clipped to the shoulder and the strobe to the back, facing away from the camera.



Life line - Photo credit Eli Wolpin

Once a balance is found, I can turn to remote lighting to augment the scene. Remote lighting can be the most complex aspect of a shoot. I primarily use remote strobes because they are more adjustable. You need to have a way to trigger a strobe, which means a direct line of sight to the sensor. I use an electrical sync cable connected to a remote sensor. The sync cable is tougher than the alternative, which is a fiberoptic connection. My remote strobe is fixed to a ball joint on a gorilla pod. This allows the strobe to be set up on uneven surfaces and directed as needed.

Good models are hard to find and worth their weight in gold. They need to be extremely patient, have impeccable trim and buoyancy, and be enthusiastic. Prior to the dive, we discuss a general plan and review hand signals. I've developed a few hand signals to direct models underwater. These are composed of three-dimensional movement and their primary light position. To communicate during a shoot, I'll turn my camera rig down onto my hands to give signals.

Prior to shooting a new scene, I'll set it up and take test shots without a model. Sometimes percolation (silty deposits falling from the ceiling) can be dislodged by the model's bubbles, so it's better to limit the time the model spends in a single location. Often, I'll put my camera down, swim to a remote strobe to fine tune positioning and power, then swim back to retest lighting. This may mean we sit in the same spot for fifteen minutes or more.

Last year, our trip got partially rained out. We arrived at the cave entrance only to find a boil emerging from the resurgence. A few days of waiting brought it down to conditions which made diving possible. I got in fewer dives than planned and only one photo dive. I had meant to capture the "shaft" which is a large vertical drop inside the cave but left disappointed and looked towards the next year. A few weeks later, while at God's Pocket Resort off Port Hardy, we hatched a plan to do a single dive at Wet Dream Cave on the way home to Vancouver. There were six of us including two rebreather divers and four on open circuit. I had been tracking water levels from river stations and it looked like there was enough reprieve from the rain that we could get a dive in.

We arrived on site and discovered it was possible to get in!

The plan was to do the shoot with three divers below me in the shaft. I sketched out a formation for the divers to hold while I attempted to capture the scene. Two of us would enter first and I would set up the remote lights and base lighting. The diver below me would have a remote strobe onboard to shine directly below on the other two divers.

Bubbles go up, which is an issue with three divers below. We planned that the deepest diver would breath normally while the other two would attempt to synchronize with him so that there would be fewer bubbles coming up. In theory, this sounds good, but in practice the time delay between bubbles coming up from the deepest diver make it extremely challenging to get the timing right.

I did my best to position myself outside the mass of bubbles and did manage to get one shot out of dozens that reasonably captures the shaft. It more than made up for the earlier trip and I look forward to going back this year to expand my photographic documentation of the cave.

Side mount tanks - Photo credit Eli Wolpin

